

Exercises for a 2-Hour React Tutorial

Setup and Warm-Up (15 minutes)

1. Set Up React:

- Ensure all students can create a new React app using `npx create-react-app` or use the provided starter files.
- Verify that everyone has the development server running (`npm start`).

Exercise 1: Create a Functional Component (20 minutes)

Task: Create a new component called `Welcome` that displays:
Welcome to React Tutorial!

- - **Steps:**
 1. Create a file `Welcome.js` in the `/src` directory.
 2. Write a functional component that renders the welcome message.
 3. Import and use `Welcome` in `App.js`.
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Exercise 2: Passing Props (25 minutes)

Task: Modify the `Welcome` component to accept a `name` prop and display:
Welcome to React Tutorial, [name]!

- - **Steps:**
 1. Update `Welcome` to use props.
 2. Pass a `name` prop when rendering `Welcome` in `App.js`.
 3. Test with different names (e.g., "Alice", "Bob").
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Exercise 3: Create a List with Components (30 minutes)

Task: Create a new component called `TodoList` that displays a list of three tasks, e.g.:

- Learn React
- Build Components
- Pass Props

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- **Steps:**
 1. Create a file `TodoList.js`.
 2. Use the `<ul>` and `<li>` tags to display the tasks.

3. Import and render `TodoList` in `App.js`.

Extension:

- Pass an array of tasks as props to `TodoList` and render the list dynamically.
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Exercise 4: Using JSX Expressions (20 minutes)

Task: Modify the `TodoList` component to include the total number of tasks at the top:

You have 3 tasks:

- Learn React
- Build Components
- Pass Props

- - **Steps:**
 1. Use a variable to store the task count (`tasks.length`).
 2. Render the count dynamically using curly braces `{}` in JSX.
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Exercise 5: Styling Components (10 minutes)

- **Task:** Add simple inline styles or CSS classes to your components.
 1. Create a `styles.css` file or use inline styling.
 2. Apply a background color and padding to the `TodoList` component.
 3. Style individual tasks in the list (e.g., bold or italic text).
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Wrap-Up and Discussion (10 minutes)

- Share and review solutions as a class.
- Discuss common errors and debugging tips.
- Suggest extending their knowledge:
 1. Learn about **state** in React.
 2. Explore **event handling** (e.g., add a button to mark tasks as completed).

This structured session introduces students to key React concepts (components, props, JSX, and styling) while providing ample hands-on practice.